

Score: / 10

Calculus I (MATH 2200)
Spring 2020
Wednesday, 02/19/2020
Quiz 4: Section 3.1, 3.2

Name (Print): _____

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Section 3.1

4 pts

1. Let $f(x) = 7x - 3$. Find $f'(x)$ using *only* the definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Using the definition of the derivative, we calculate:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(7(x+h) - 3) - (7x - 3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{7x + 7h - 3 - 7x + 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{7h}{h} \\ &= \lim_{h \rightarrow 0} 7 = 7. \end{aligned}$$

4 pts

2. Let $f(x) = \frac{1}{x-5}$. Find $f'(x)$ using *only* the definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Using the definition of the derivative, we calculate:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{1}{(x+h)-5} - \frac{1}{x-5} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{1}{x+h-5} - \frac{1}{x-5} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{(x-5)}{(x+h-5)(x-5)} - \frac{(x+h-5)}{(x+h-5)(x-5)} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{(x-5) - (x+h-5)}{(x+h-5)(x-5)} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{-h}{(x+h-5)(x-5)} \right) \\ &= \lim_{h \rightarrow 0} \frac{-1}{(x+h-5)(x-5)} = \frac{-1}{(x-5)^2} \end{aligned}$$

Section 3.2

2 pts

1. The graph of a function f is given below. Sketch a graph of f' using this information.

